

Last login: Wed Jun 17 15:33:30 on ttys000
psql [1]
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# SET ROLE alice;
SET
tpch=> SELECT count(*)
FROM lineitem;
WARNING:  Log 1: Planner Input:
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_status" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCI true
    :securityQuals (
      {SUBLINK
        :subLinkType 0
        :subLinkId 0
        :testexpr <>
        :operName <>
      }
    )
    :subSelect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks false
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTTRY
            :alias
              {ALIAS
                :aliasname s
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname s
                :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
              }
            :rtekind 0
            :relid 16541
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCI true
            :securityQuals <>
          }
        )
        :rtepermissions (
          {RTEPERMISSIONINFO
            :relid 16541
            :inh true
            :requiredPerms 2
            :checkAsUser 0
            :selectedCols (b 13)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    )
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
)
:funcexpr
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
```

```

        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -24 3 0 0 0 0 0 0 ]
        }
    }
    :location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
)
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
rtindex 1
}
)
}
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 28
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 7
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 29
}
)

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
)
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: {PATH
```

```

:pathtype 339
:parent_relds (b 1)
:pathtarget
  {PATHTARGET
    :exprs <>
    :sortgrouprefs ()
    :cost.startup 0
    :cost.per_tuple 0
    :width 0
    :has_volatile_expr 0
  }
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 8196
:disabled_nodes 0
:startup_cost 0
:total_cost 343
:pathkeys <>
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

DETAIL: QUERY

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtekind 0
    :relid 16566
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 16
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 0
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggssplit 0
        :aggno -1
        :aggtransno -1
        :location 7
      }
    :resno 1
    :resname count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>

```

```

:stmt_location 0
:stmt_len 29
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocons
tant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relds (b) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left
_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :
hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0))
WARNING: max_parallel_hazard_walker: Param paramid=0 in safe_list=1 * reason=outer reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocon
stant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relds (b) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :lef
t_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false
:hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :left_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfil
ter <> :aggstar true :aggvariadic false :aggkind n :aggsorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfil
ter <> :aggstar true :aggvariadic false :aggkind n :aggsorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: {AGGPATH
:path.pathtype 365
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 133849.57018301616
:path.total_cost 133849.58018301617
:path.pathkeys <>
:subpath
{GATHERPATH
:path.pathtype 368
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 2
:path.disabled_nodes 0
:path.startup_cost 133849.31333333335
:path.total_cost 133849.52333333337
:path.pathkeys <>
:subpath
{AGGPATH
:path.pathtype 365
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 132849.31333333335
:path.total_cost 132849.32333333336
:path.pathkeys <>
:subpath
{PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware true
:parallel_safe true
:parallel_workers 2
:rows 1250192
:disabled_nodes 0
:startup_cost 0
:total_cost 129723.83333333334
:pathkeys <>
}
:aggstrategy 0
:aggsplit 6
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}
:single_copy false
:num_workers 2
}
:aggstrategy 0
:aggsplit 9
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:
DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded true
:jitFlags 25
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 133849.57018301616
:plan.total_cost 133849.58018301617
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 0
        :varattnosyn 0
        :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 9
:aggno 0
:aggtransno 0
:location 7
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{GATHER
:plan.disabled_nodes 0
:plan.startup_cost 133849.31333333335
:plan.total_cost 133849.52333333337
:plan.plan_rows 2
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 132849.31333333335
:plan.total_cost 132849.32333333336
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location 7
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1

```

```

:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.leftright <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanreldid 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numcols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:num_workers 2
:rescan_param 1
:single_copy false
:invisible false
:initParam (b 0)
}
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe true
:setParam (i 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 0.041849682772083945
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:aggstrategy 0
:aggssplit 9
:numcols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines_status" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:reldid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}

```

```

:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:unprunableRelids (b 1 2)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 2
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16566 16541)
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location 0
:stmt_len 29
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: (PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 132849.31333333335
:plan.total_cost 132849.32333333336
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n

```

```

:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggn 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location -1
    }
    :resno 13
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 14
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 14
    :location -1
    }
    :resno 14
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location -1
    }
    :resno 15
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location -1
    }
    :resno 16
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :scan.plan.qual (
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 16
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b 0 1)
    :scan.plan.allParam (b 0 1)
    :scan.scanrelid 1
    )
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0 1)
    :plan.allParam (b 0 1)
    :aggstrategy 0
    :aggssplit 0
    :numCols 0
    :grpCollidx <>
    :grpOperators <>
    :grpCollations <>
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    )
    :partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtkind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref

```

```

        {ALIAS
        :aliasname s
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
        }
:rtelid 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
    }
)
:unprunableRelids (b 1 2)
:permInfos (
    {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 13)
    :insertedCols (b)
    :updatedCols (b)
    }
)
:resultRelations <>
:appendRelations <>
:subplans (
    {SEQSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 343
    :scan.plan.plan_rows 8196
    :scan.plan.plan_width 0
    :scan.plan.parallel_aware false
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 4
    :scan.plan.targetlist <>
    :scan.plan.qual (
        {OPEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
            {VAR
            :varno 2
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 6
            :location -1
            }
            {CONST
            :consttype 701
            :consttypmod -1
            :constcollid 0
            :constlen 8
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
            }
        )
        :location -1
        }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b)
    :scan.plan.allParam (b)
    :scan.scanrelid 2
    }
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
    {AGG
    :plan.disabled_nodes 0
    :plan.startup_cost 132849.31333333335
    :plan.total_cost 132849.32333333336
    :plan.plan_rows 1
    :plan.plan_width 8
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 2
    :plan.targetlist (
        {TARGETENTRY
        :expr
        {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 20
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        }
        }
    )
    }
}

```

```

:agglevelsup 0
:aggssplit 6
:aggnr 0
:aggransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13

```

```

        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location -1
    }
}
:resno 13
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
}
:resno 15
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
}
:resno 16
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numcols 0
:grpColldx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtkind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS

```

```

        :aliasname s
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
}
)
:unprunableRelids (b 1 2)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:varstype 701
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.plan.scanrelid 2
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

count
-----
6001215
(1 row)

tpch=>

```